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Raphaël Fournier[1*], Paul Downing[2] & Richard Ramsey[1]

[1]Neural Control of Movement Laboratory, Department of Health Sciences and Technology, ETH Zurich

[2]Name, department

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Introduction

The ability to quickly and correctly recognize a familiar face is important in our everyday life. Not only does this ability allow us to anticipate another person's behaviour, but this recognition will influence our own behaviour as well. People will behave differently in function of the identity of the person they are interacting with. Most of the previous research focusing on face recognition has used manifest measures (i.e. response times and accuracy) to assess one's ability to recognize faces. However, less is known about the computational processes that support face recognition. We investigated how different types of face familiarity affected the latent variables estimated by evidence accumulation models (EAMs; Ratcliff and Rouder (1998)). This computational approach has provided new insights on the dynamics of the decision process supporting face recognition.

The face recognition process has been intensely investigated over the last 50 years (Johnston and Edmonds 2009; Natu and O'Toole 2011). One of the most influential models describing the face recognition process is the functional model of face recognition proposed by Bruce and Young (1986). This model advances that visual recognition happens when the encoded structural representation of a face matches the stored structural code of the same face. These structural codes are contained in the face recognition units (FRUs) which have access to person identity nodes that contains identity-specific semantic information. Based on neuroimaging findings, researchers have adapted this model to relate its different components to specific brain areas (Gobbini and Haxby 2006, 2007; Haxby, Hoffman, and Gobbini 2000). This general model of the distributed neural system for familiar face recognition incorporates two distinct systems. The core system, composed by the inferior occipital gyrus (occipital face area, OFA), the inferior fusiform gyrus (fusiform face area, FFA) and the posterior superior temporal sulcus (pSTS), is dedicated to modality-specific visual processing of faces. The core system is involved in the recognition of faces established on visual features only. The extended system participates in the encoding of the person knowledge and is mediated by a broader network of brains areas (Duchaine and Yovel 2015; Gobbini and Haxby 2007; Haxby, Hoffman, and Gobbini 2000; Ishai 2008; Kanwisher and Barton 2011). The extended system support the recognition of faces for which prior semantic knowledge have been stored, like famous faces (add ref.) or personally familiar faces (add ref.), by mediating the retrieval of identity-specific information from memory.

Previous studies have shown that the association of visual familiarity and identity-specific information was also reflected on manifest variables (i.e response times and accuracy) in tasks assessing face recognition ability. Using face memory paradigms, a large number of studies have shown that the recognition of famous faces or personnaly familiar faces was faster and more accurate than visually familiar faces (ref). Using face matching tasks, similar conclusions have been drawn: people match faster famous faces than visually familiar faces (ref).

*Send correspondence to: Raphaël Fournier, rfournier@ethz.ch.
Author notes go here.

• Person knowledge familiarity: (**akan2023?**; **schwaninger2002?**; **clutterbuck2005?**; **read?**; **jenkins2011?**; **johnston2009?**)

• Other possible refs: (e.g., Cloutier, Kelley, & Heatherton, 2011; Dubois et al., 1999; Gobbini & Haxby, 2006; Guntupalli, Wheeler, & Gobbini, 2017; Kosaka et al., 2003; Natu & O'Toole, 2015; Rossion, Schiltz, Robaye, Pirenne, & Crommelinck, 2001).

For the last 50 years, research on face perception and face recognition has been singularly dominated by the separate analysis of manifest variables *(i.e. response times and accuracy; ref) or a transformation of the accuracy score (d'; ref).* The analysis of theses variables has outlined the understanding framework of face perception and face recognition. The ability to recognize a face is defined in terms of response times and accuracy, separately. Inferences are then drawn from the difference in accuracy and response times across different stimuli (ex: famous faces vs visually familiar faces). However, defining an ability based on performance alone narrows the inferences that could be done regarding the studied ability. In some extreme cases, it can even lead to misinformed inferences regarding a specific effect (Ratcliff, Thapar, and McKoon 2001). ***Here need a link sentence.*** In human neuroscience and social cognition research, there is a lack of application these computational approaches (Parker and Ramsey, n.d.; Forstmann and Turner 2024). Using computational approaches, (why important?)

Evidence accumulation models (EAM) are one approach to investigate the computational processes involved in face recognition. For the last two decades, the application of these models has defined the computational mechanisms involved in several cognitive processes (for a review, see Ratcliff et al. (2016)). While a variety of different evidence accumlation models exists, they all share the same aim: translating accuracy and the distribution of RTs for correct and incorrect responses into estimates of latent variables assumed to underlie the decision-making process (Brown and Heathcote 2008; Donkin, Brown, and Heathcote 2011). Drift rate is the rate at which evidence in favor of a decision accumulates over time, reflecting the signal-to-noise ratio provided by the stimulus. Threshold represents the amount of accumulated evidence needed to trigger the decision, reflecting the response caution towards a specific response. The starting point is the amount of evidence towards a response before the accumulation of evidence begins, reflecting the *a priori* bias towards a response. Finally, the non-decision time is

• Figure 1: LBA figure (I know which one)

• Latent psychological variables of interest: drift rate (evidence of accumulation) and threshold (response caution).

• Past success in investigating psychology (mostly decision-making,), some applications in human social cognition

• Now: Increased application of these models to investigate other cognitive process, social cognition…

Four experiments were conducted to investigate how familiarity affects the decision-making process of face recognition using evidence-accumulation models, more specifically the LBA model. In Experiment 1 and2, face familiarity was experimentally induced. Each participants underwent a familiarization part where they got visually familiar with 30 previously unseen faces, In the recognition part, participants had to choose if the presented face has been previously seen during the familiarization part (learned faces) or not (novel faces). Experiment 2 was a replication of Experiment 1 conducted on a online platform for data collection (https://pavlovia.org). In Experiment 3 and 4, familiarity was modulated using the a prior familiarity of the participants towards celebrities faces. The first part of the experiment was a rating part to assess the familiarity level of the participants towards 60 famous face. In the recognition part, the 30 most familiar faces were mixed to novel faces. For all experiments, we hypothesized that either drift rate and/or threshold would be affected by the face familiarity manipulation. The direction of the effects would be similar when comparing Experiment 1 and 2 and when comparing Experiment 3 and 4.

Experiment 1

Method and Materials

Pre-registration and open science

The research question, hypotheses, experimental design, planned analysis, and exclusion criteria were preregistered (link). All raw data, stimuli, data wrangling and analysis code are available on the open science framework (see link).

Results

Discussion

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